AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-08-29 | Entry window 29 August 2026 | 8 records screened in scope

8 RECORDS IN SCOPE (22 EXCLUDED)	**7** SUBTOPIC CLUSTERS OCCUPIED (OF 10)	**4** REVIEWS / POSITION STATEMENTS	**5** EFFECT ESTIMATES FROM 1 STUDY	**2** RECORDS FROM SUB-SAHARAN AFRICA

Signal of the day

Thyroid function joins the list of endpoints with a coherent air-pollution dose–response — and PM_{10}, not $PM_{2.5}$, carries the strongest association.

Park et al. (*J Endocrinol Invest*) screened **16,193 Korean adults** with TSH in the assayable 0.1–10.0 mIU/L range against CMAQ-modelled annual $PM_{2.5}$, PM_{10}, NO_2, SO_2 and CO. Every pollutant is positively associated with TSH and negatively with free thyroxine ($P < 0.001$), and for the combined **high-TSH/low-FT4 phenotype** the adjusted odds ratios per IQR order as **PM_{10} 1.55 (95% CI 1.42–1.69)** $>$ SO_2 1.38 (1.28–1.49) $>$ NO_2 1.36 (1.23–1.50) $>$ **$PM_{2.5}$ 1.25 (1.17–1.34)** $>$ CO 1.23 (1.15–1.31), with stronger effects in women. That PM_{10} outranks $PM_{2.5}$ is worth sitting with: five collinear pollutants from a single CMAQ field, entered in separate single-pollutant models, cannot separate coarse-fraction mechanism from correlation structure — but if the coarse fraction really is doing the work, it is the fraction most low-cost optical networks measure worst, because PM_{10} sizing in a nephelometric unit is the least reliable channel it has. Cross-sectional design and a single-hospital screening population cap the inference at hypothesis strength.

What changed today

- **A thin day, and half of it is synthesis.** Eight records over seven clusters, of which **four are reviews or position statements**. No cluster holds more than two.
- **Sub-Saharan Africa appears twice after an eight-issue absence.** A West Rand time-activity survey (330 households near gold-mine tailings) and a meningitis-belt dust review. The time-activity paper is the more useful for exposure work: it reports indoor fractions **roughly 10 percentage points below the USEPA Exposure Factors Handbook values** that most exposure models silently import, which biases indoor–outdoor apportionment for any African deployment using default time budgets.
- **Microplastics enter through a particle-toxicology route.** Chen et al. (*Part Fibre Toxicol*) quantify placental microplastic burden by pyrolysis-GC-MS and stratify transcriptome, proteome and metabolome by it — the first record in this watch to treat placental MP burden as a graded internal dose rather than a presence/absence finding.
- **A cautionary machine-learning result.** Yu et al. (*Indoor Air*) predict seven self-reported symptoms in Milan dormitories from continuous environmental quality data: **mean AUC 0.595 (0.398–0.705)**, only four models above 0.6, from 74 occupant samples and 131 candidate predictors. The paper's own observation is the interesting one — dominant predictors were **outdoor statistics and distributional shape features, not mean indoor concentrations**. Mean-value summaries of a sensor time series discard most of the health-relevant information.
- **Two records that are mostly agenda-setting.** The SynAir-G statement on children's indoor air quality argues for chemical$\times$biological synergy as the policy-relevant unit of exposure; the poultry-worker airway review restates a well-established organic-dust hazard without new quantification.
- **Rejected but worth naming.** Two directly on-topic Elsevier deposits — a hybrid AQI forecasting framework (*Atmos Pollut Res*) and an operational ragweed-pollen forecast evaluation (*Environ Pollut*) — arrived with no abstract in any indexed source and are carried to the retry list rather than summarised from titles.

Corpus at a glance

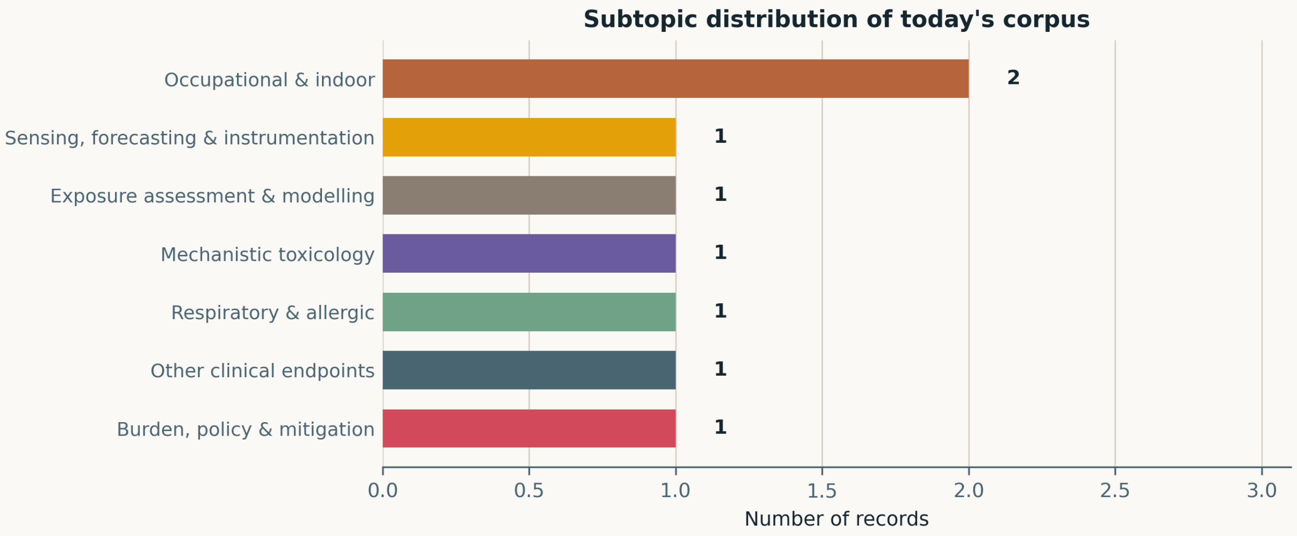

Fig. Records per subtopic cluster: 8 records over **seven of ten** clusters, with *Occupational & indoor* the only cluster holding 2. *Cardiovascular & metabolic*, *Neuro / mental health* and *Reproductive & developmental* are empty — although the placental microplastics record is reproductive in substance and sits under *Mechanistic toxicology* because its contribution is molecular rather than obstetric. A flat distribution like this one carries no signal about the field; it is what a low-volume deposit day looks like.

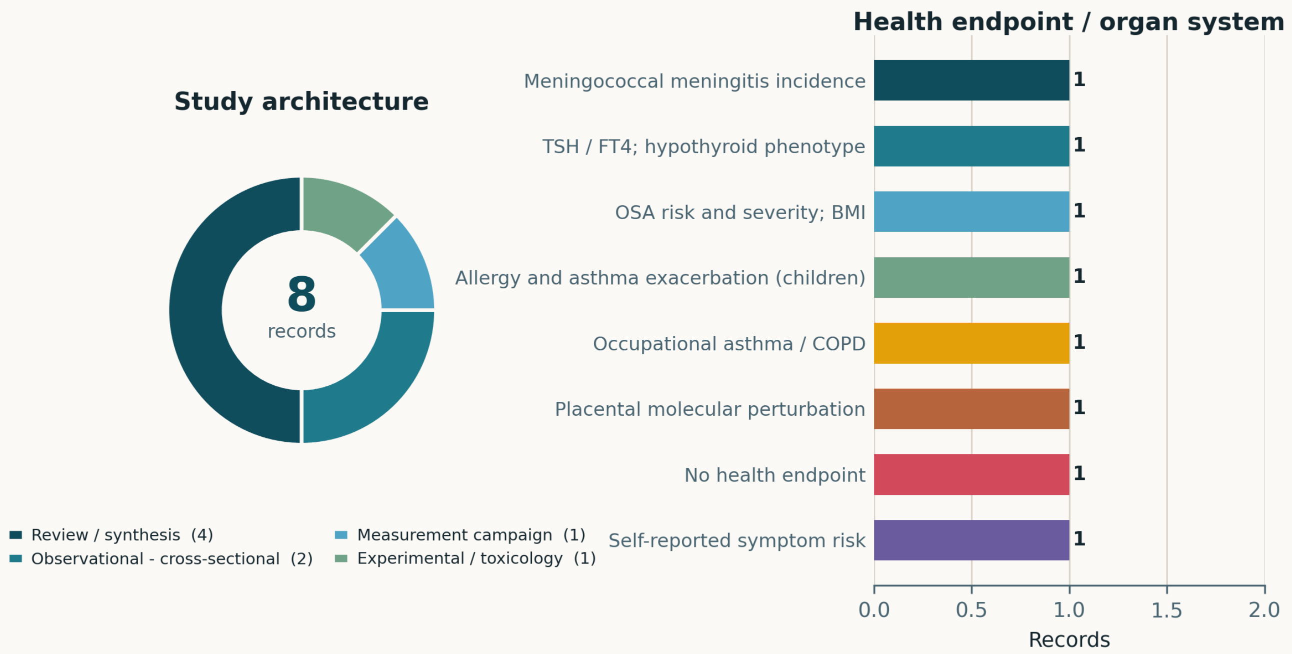

Fig. Left — study architecture, and the dominant fact of this issue: ***Review / synthesis* is 4 of 8**, against *Observational — cross-sectional* 2, *Experimental / toxicology* 1 and *Measurement campaign* 1. A day with more synthesis than primary data is a deposit-calendar artefact, not a statement about research direction. Right — health endpoint: one record (the West Rand time-activity survey) carries none, and no endpoint repeats.

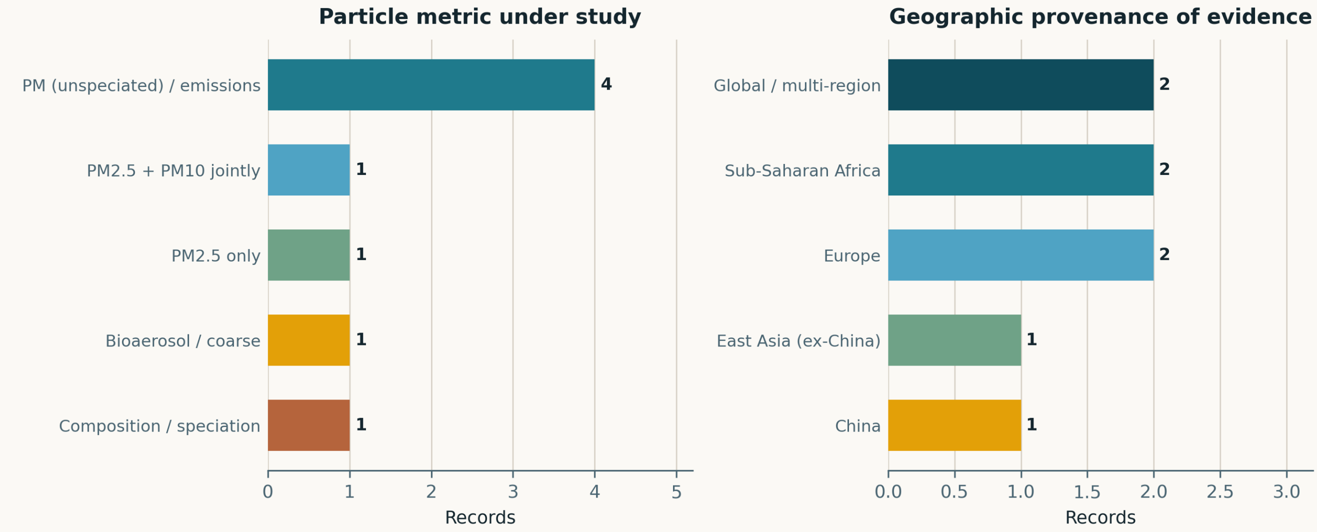

Fig. Left — particle metric. *PM (unspeciated) / emissions* 4 dominates, because half the corpus characterises a mixed or organic dust rather than a regulated mass fraction; one each for *$PM_{2.5}$ only*, *$PM_{2.5}$ + PM_{10} jointly*, *composition / speciation* (placental polymer burden) and *bioaerosol / coarse*. Right — geography: Europe 2, **Sub-Saharan Africa 2**, multi-region 2, China 1, East Asia ex-China 1. The African pair ends an eight-issue absence running from 21 August; the last previous African record was the Addis Ababa study of 20 August.

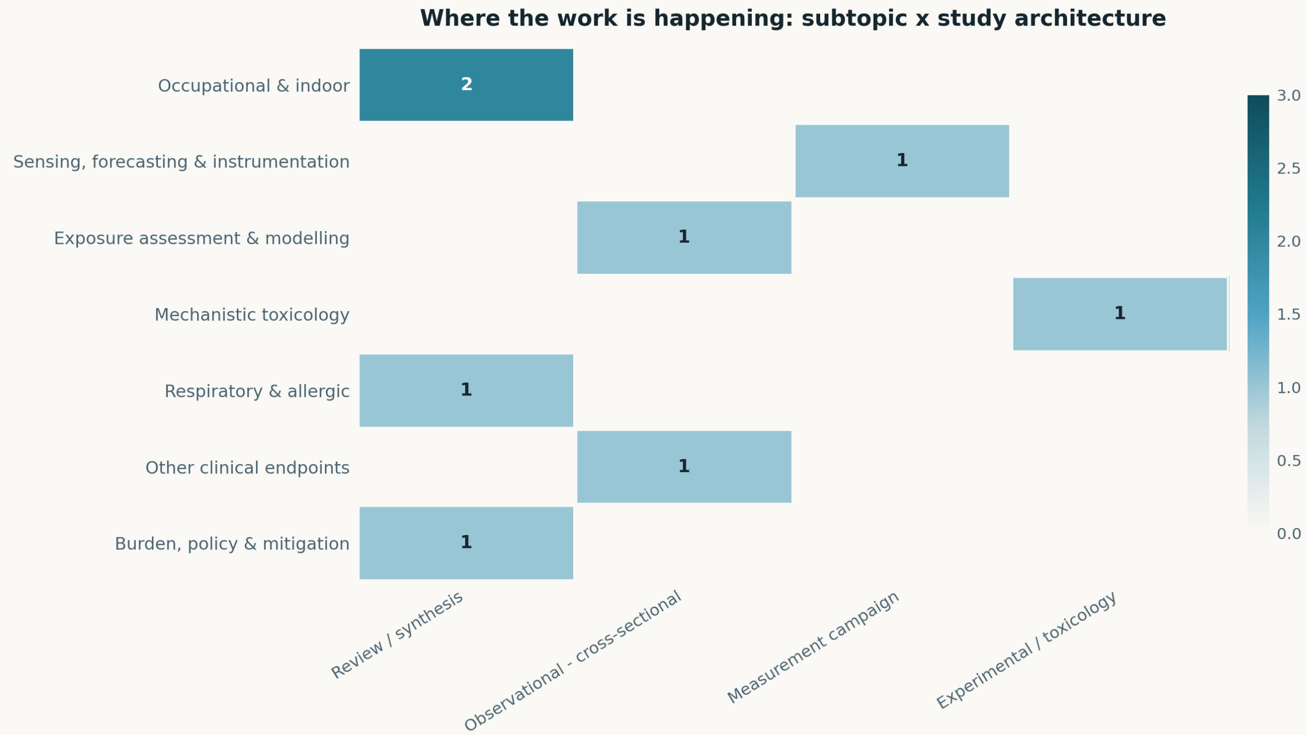

Fig. Subtopic × architecture cross-tabulation. With 8 records over seven clusters and four architectures, **only one cell holds 2** — *Occupational & indoor* × *Review / synthesis* — and the other six occupied cells hold 1. There is no structure to read here; the panel is carried for continuity of the series.

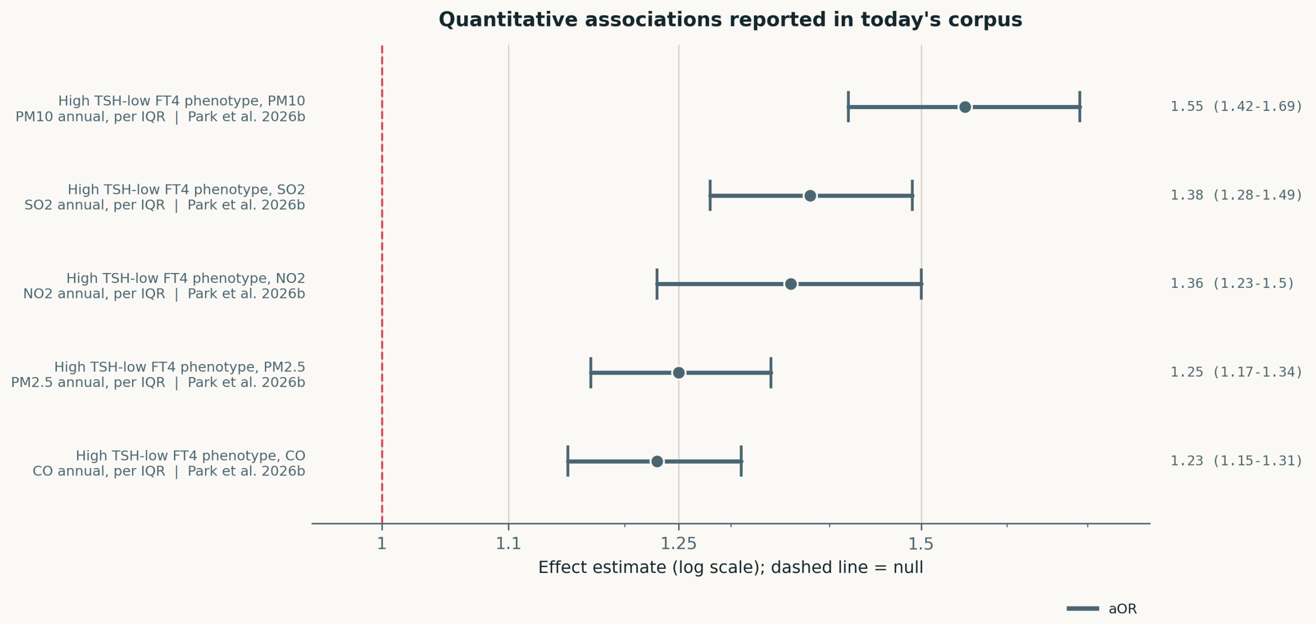

Fig. Five estimates, **all from a single study** (Park et al.), all for the same outcome and all on a common per-IQR contrast, so this is one of the few panels in this watch where a pollutant-versus-pollutant reading is internally consistent. It is *not* a potency comparison in any causal sense: the five exposures come from one CMAQ field, are mutually collinear, and were entered in separate single-pollutant models, so the ordering ($PM_{10} > SO_2 > NO_2 > PM_{2.5} > CO$) partly reflects each pollutant's IQR width and its correlation with the others rather than an independent effect. The intervals are narrow because n is large, not because confounding is small.

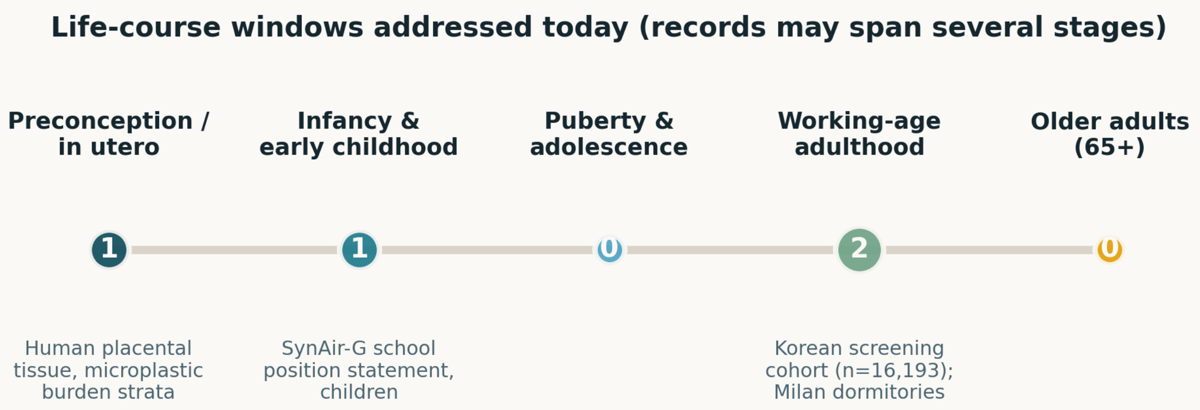

Fig. Life-course windows. **Four records across three of five windows**: the in-utero cell holds the placental microplastics study, the childhood cell the SynAir-G schools statement, and the working-age cell both the Korean screening cohort and the Milan dormitory panel. Puberty/adolescence and 65+ are empty.

Paper-level digest

Ordered by cluster; the 29 August entry window only.

Sensing, forecasting & instrumentation 1 record

Yu et al. 2026b Indoor Air

Machine learning-based prediction of occupant symptom risk using indoor and outdoor environmental quality data in university dormitories

Seven monitoring periods across two Milan student residences, May 2024 to June 2025, pairing continuous indoor and outdoor environmental quality data with occupant symptom surveys. From **74 occupant samples** and 131 candidate predictors (reduced per symptom to 3–40 by permutation importance), seven binary symptom classifiers were fitted under leave-one-period-out cross-validation. Performance is weak and honestly reported: **mean AUC 0.595, range 0.398–0.705**, four of seven above 0.6; the higher combined accuracy figures appear only after dropping the two least specific symptoms. With $n = 74$ against 131 candidate predictors the feature-selection step is almost certainly overfitting even under LOPO-CV. The transferable finding is the feature ranking: **outdoor statistics and distributional shape features beat mean indoor concentrations**, which argues directly against summarising a sensor time series by its mean.

doi: 10.1155/ina/8230467

Exposure assessment & modelling 1 record

Kalumbi et al. 2026 J Expo Sci Environ Epidemiol

Age-specific time-spent patterns among residents living near gold mine tailings storage facilities in the West Rand, South Africa

Cross-sectional structured-questionnaire survey of **330 households in Kagiso**, West Rand, resolving indoor and outdoor time across five age categories with Kruskal–Wallis tests and Bonferroni-adjusted Dunn's comparisons. Indoor time runs **66.6–84.8% of a weekday and 76.0–82.7% of a weekend day** (18.28±3.02 h weekdays, 19.00±1.99 h weekends); young children and older adults are indoors most, and weekend indoor time rises significantly for children, adolescents and adults but not for the two extreme age groups. The headline for exposure modelling is that indoor time is **nearly 10 percentage points below USEPA Exposure Factors Handbook defaults**. Self-reported recall with no diary or wearable validation, and a single district, so the magnitude is soft — the direction of the bias in imported default time budgets is the usable result.

doi: 10.1038/s41370-026-00972-6

Mechanistic toxicology 1 record

Chen et al. 2026j Part Fibre Toxicol

Multi-omics integration reveals human placental molecular signatures associated with microplastics exposure

Placental microplastic burden quantified by **pyrolysis-gas chromatography–mass spectrometry**, samples stratified into high- and low-burden groups, then transcriptomic, proteomic and metabolomic profiling integrated through the DIABLO framework with network analysis for regulatory hubs. High-burden placentae show multi-layer molecular perturbation centred on immune dysregulation. The design's strength is that MP burden is treated as a *graded internal dose* rather than a detection/non-detection flag, which separates it from most human-tissue microplastics work. Its limits are the usual ones for cross-sectional human tissue: burden is measured once at delivery, exposure timing is unknown, the omics contrasts are correlational, no sample size appears in the abstract, and Py-GC-MS quantifies polymer *mass* without particle size or count — so the dose metric is mass, not number or surface area.

doi: 10.1186/s12989-026-00697-2

Occupational & indoor 2 records

Ramirez et al. 2026 Curr Allergy Asthma Rep

Airway diseases and occupational exposures in poultry industry workers: a comprehensive review

Review of occupational asthma and chronic obstructive lung disease in poultry workers, attributing risk to the barn's mixed aerosol — feather and feed dust, mites, moulds, ammonia and bacterial endotoxin — acting through airway inflammation and hypersensitivity, with a symptom spectrum from wheeze and chest tightness to chronic cough and dyspnoea. Emphasis falls on early identification through occupational history-taking and on barn control practices. **No prevalence figures, no exposure–response estimates and no search or inclusion strategy** are reported, so this is a clinical orientation piece rather than an evidence synthesis. Its relevance here is the exposure architecture: an occupational aerosol whose health-relevant fraction is defined by biological composition, which a mass-only monitor cannot resolve at all.

doi: 10.1007/s11882-026-01289-y

Siemens et al. 2026 Indoor Air

A SynAir-G position statement for the improvement of indoor air quality for children

Position statement from the EU-funded SynAir-G project (18 partners, 11 countries) within the seven-project IDEAL Cluster, arguing that indoor air has been under-regulated relative to outdoor air despite most exposure occurring indoors, and that the policy-relevant unit is the **synergistic interaction between chemical and biological pollutants** rather than any single-pollutant concentration. Schools and children are the target setting. As a consensus document it presents no new measurements, no systematic evidence appraisal and no quantified synergy, so the central claim is asserted rather than demonstrated. The instrumentation implication is nonetheless concrete: a synergy framing requires co-located chemical and bioaerosol sensing at classroom scale, which no current low-cost platform delivers.

doi: 10.1155/ina/9690417

Respiratory & allergic 1 record

Silva et al. 2026 Front Sleep

Shared mechanisms linking $PM_{2.5}$ exposure, obstructive sleep apnea and obesity

Narrative review of the three-way relationship between chronic $PM_{2.5}$ exposure, obstructive sleep apnea and obesity, proposing systemic inflammation, oxidative stress, neuroendocrine dysregulation and gut microbiota alteration as shared pathways, and noting that intermittent hypoxia and sleep fragmentation feed back into weight gain. The mechanistic overlap is plausible and the review is explicit that it is hypothesis-organising, but **no effect estimates, no study inventory and no appraisal of confounding by adiposity** appear — and adiposity confounds both the exposure (residential location) and the outcome. Read alongside the CHARLS sleep-duration result carried on 28 August, the practical gap is identical in both: no objective sleep measurement anywhere in this literature.

doi: 10.3389/frsle.2026.1918405

Other clinical endpoints 1 record

Park et al. 2026b J Endocrinol Invest

Association between annual air pollutant exposure and thyroid function: a retrospective cross-sectional study

16,193 adults (8,812 male, 7,381 female; mean age 53.8±13.1 y) screened at Seoul National University with TSH within 0.1–10.0 mIU/L, against CMAQ-estimated annual $PM_{2.5}$, PM_{10}, NO_2, SO_2 and CO. All five pollutants associate positively with TSH and negatively with free thyroxine ($P < 0.001$), with significant non-linearity for $PM_{2.5}$ and NO_2 against log TSH. For the combined high-TSH/low-FT4 phenotype the per-IQR adjusted odds ratios run **PM_{10} 1.55 (1.42–1.69)**, SO_2 1.38 (1.28–1.49), NO_2 1.36 (1.23–1.50), **$PM_{2.5}$ 1.25 (1.17–1.34)** and CO 1.23 (1.15–1.31), stronger among women. The binding limitations: exposure is a modelled annual field at residence with no personal component, the design is cross-sectional so temporality is unresolved, and the five pollutants are collinear by construction — the ordering must not be read as differential potency.

doi: 10.1007/s40618-026-02965-6

Burden, policy & mitigation 1 record

Cliff et al. 2026 GeoHealth

How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt?

Narrative review over the 26-country African meningitis belt, where outbreaks cluster strongly in the dry season, examining dust, wind speed, temperature, rainfall and land cover as candidate drivers. **Atmospheric dust and wind speed carry the strongest statistical association and the greatest predictive probability** of the variables examined, with mucosal damage from inhaled mineral dust the standing mechanistic hypothesis. The review's own central caveat is the right one: these climatic variables are severely multicollinear within the dry season, so attributing incidence to dust rather than to temperature or humidity is not identified by the observational record. Carried here because it is one of the few live health endpoints where the *coarse* mineral fraction, not $PM_{2.5}$, is the hypothesised agent — and where satellite dust products substitute for a ground network that does not exist.

doi: 10.1029/2025gh001749

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Silva et al.	Shared mechanisms linking $PM_{2.5}$ exposure, obstructive sleep apnea and obesity	*Front Sleep*	Mechanistic review	Chronic ambient $PM_{2.5}$	Global / multi-region
Mechanistic toxicology					
Chen et al.	Multi-omics integration reveals human placental molecular signatures associated with microplastics exposure	*Part Fibre Toxicol*	Human tissue multi-omics (transcriptome/proteome/metabolome)	Placental microplastic mass burden by Py-GC-MS, high vs low strata	China
Occupational & indoor					
Ramirez et al.	Airway diseases and occupational exposures in poultry industry workers: a comprehensive review	*Curr Allergy Asthma Rep*	Comprehensive review	Poultry-barn organic dust: feather, mite, feed dust, mould, endotoxin, NH3	Global / multi-region
Siemens et al.	A SynAir-G position statement for the improvement of indoor air quality for children	*Indoor Air*	Consensus position statement	Indoor chemical and biological pollutants, bioaerosols, synergistic mixtures	Europe
Sensing, forecasting & instrumentation					
Yu et al.	Machine learning-based prediction of occupant symptom risk using indoor and outdoor environmental quality data in university dormitories	*Indoor Air*	Continuous IEQ monitoring + supervised classification	Five indoor and five outdoor environmental parameters incl. PM; distributional shape features	Italy
Exposure assessment & modelling					
Kalumbi et al.	Age-specific time-spent patterns among residents living near gold mine tailings storage facilities in the West Rand, South Africa	*J Expo Sci Environ Epidemiol*	Cross-sectional time-activity survey	None measured; indoor/outdoor time fractions as exposure-weighting inputs	South Africa
Burden, policy & mitigation					
Cliff et al.	How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt? A narrative literature review	*GeoHealth*	Narrative review	Atmospheric mineral dust load and wind speed	Sub-Saharan Africa
Other clinical endpoints					
Park et al.	Association between annual air pollutant exposure and thyroid function: a retrospective cross-sectional study	*J Endocrinol Invest*	Retrospective cross-sectional (health screening)	Annual mean $PM_{2.5}$ and PM_{10} from CMAQ, per IQR	South Korea

Provenance and method

Entry window **2026-08-29 to 2026-08-29**, a single day of deposits. **Sources.** PubMed E-utilities on `[Date - Entry]`, run as two separate queries (health axis 20, instrumentation axis 3). Europe PMC `CREATION_DATE` returned 0. Crossref by-ISSN returned 79 deposits across the harvester's two-day window; because that window opened at `last_entry_date + 1` while the 28 August issue was still being written, **57 of the 79 carry a Crossref creation date of 28 August and were screened in the previous issue** — they are excluded here rather than re-screened, so this issue's Crossref pool is the 22 deposits created on 29 August. OpenAlex remains broken for entry-date windowing; arXiv contributed nothing in window. **Screening. 30 unique** candidates after PMID/DOI collapse and removal of records already in `seen.json` (15 of the 23 PubMed hits were already published in the 28 August issue), **8 retained and 22 rejected** with reason logged to `state/rejected.jsonl`. Rejections were dominated by non-particle environmental science (*Sci Total Environ* soil, water and carbon-flux deposits plus four retraction notices), non-airborne chemical exposure (bisphenol A, 2,4-DTBP, endocrine-disruptor mixtures) and *Meas Sci Technol* non-atmospheric metrology. **Two in-scope deposits** — an AQI forecasting framework and a ragweed-pollen operational forecast — had no retrievable abstract and were carried to the retry list rather than summarised. **Effect estimates.** Five estimates, all from Park et al.; the figure caption states why the between-pollutant ordering is not a potency comparison. **Pipeline changes made this run.** Six design labels and *Italy* were added to `plots.py` in the same edit that wrote them onto records. **Labels.** Subtopic, design, particle-metric and geography are single-label by dominant contribution — a judgement, not a controlled vocabulary. All 8 DOIs passed `check_dois.py` (0 fail, 0 warn) before build.